

SIMONE MARIA GIANCOLA

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PROFILE

Background in statistics, computer science, and probability. Passionate about the interplay of rigorous research and insights from physics.

CURRENT

PhD researcher 2025 - present
Orsay institute of mathematics (IMO)

- Advisors: C. Giraud, N. Verzelen (INRAE Montpellier)
- Statistical to computational gaps, low-degree method, statistical physics

EDUCATION

Orsay institute of mathematics (IMO), Univ. Paris-Saclay 2024 – 2025
M2 probability and statistics

- mention *Très bien*
- thesis “Computational lower bounds via almost orthonormal polynomials”; advisors A. Carpentier, C. Giraud, N. Verzelen

Bocconi University 2021 – 2024
MS data science

- GPA 29 / 30 | 110 *Cum Laude* / 110
- thesis: “A roadmap to message passing methods for inference on mixed generalized linear models, with emphasis on mixed noiseless phase retrieval”; advisor C. Lucibello

Bocconi University 2018 - 2021
BS economics, management and computer science

- GPA 29.18 / 30 | 110 *Cum Laude* / 110
- semester abroad at Arizona State University (ASU) spring 2021

PAPERS

Carpentier, **Giancola**, Giraud, and Verzelen. *Low-degree lower bounds via almost orthonormal bases*. [arXiv:2509.09353 \[stat.ML\]](https://arxiv.org/abs/2509.09353), 2025.

Richtárik, **Giancola**, Lubczyk, and Yadav. *Local curvature descent: squeezing more curvature out of standard and Polyak gradient descent*. In: [NeurIPS25](#).

EXPERIENCE

Orsay institute of mathematics(IMO) 2025
with C. Giraud, N. Verzelen (INRAE Montpellier), A. Carpentier (Potsdam)

- Low-degree method, statistical-to-computational gaps

KAUST 2024
with P. Richtárik

- Optimization, conditioned gradient descent methods, machine learning

ENS Paris 2023
with B. Loureiro

- Neural network theory, stochastic gradient descent, gradient flow

IST Austria 2023
with M. Mondelli

- Statistical-to-computational gaps, information theory, message passing algorithms

TEACHING

Statistical modelling; TA, Master

ENSTA, 2026

Statistical tests for biology & Ecology and statistics; TA, Bachelor

IMO Orsay, 2026-2027

Analysis & Graph theory; TA, Bachelor

IUT d'Orsay, 2026

SERVICE

Organizer 2026-2028 Seminar for M2StatML, IMO Orsay

Reviewer NeurIPS 2024, ICLR 2025, NeurIPS 2026.

Mentor Prison math project (PMP); LeadTheFuture.

TALKS

Potsdam group seminar

2026

SuPerGRandMa weekly seminar, Orsay

2026

Séminaire des doctorants, U. Paris Cité

2026

Potsdam group seminar

2025

CS331 Stochastic gradient descent methods, KAUST

2024

Séminaire des doctorants, ENS Paris

2023

IST Austria group seminar

2023

DISTINCTIONS

Selected for the 12th Heidelberg laureate forum (HLF).

Scholarships: OeAD Austria ref. num. MPC-2023-01128; first Italian school in geometric deep learning.

SKILLS

Advanced Python, $\text{\LaTeX}$;

Basic Julia, Git, Unix, R, SQL, Matlab, C++, Keras, TensorFlow.

LANGUAGES

English (proficient); French (intermediate); Spanish (basic); Italian (native).